

with an NTC thermistor network, a multi-cell (2S/3S/4S) Type-C charging module, a 4-cell 18650 battery holder, the INA219 current sensor, and a few passive components. The design integrates voltage sensing, temperature monitoring, and current measurement into a compact microcontroller-based system.

In this circuit, the voltage of each cell is measured individually to estimate its SoC. A resistor divider network is used to scale down the cell voltage to a level suitable for the microcontroller's analogue input. Although this is a concept design, higher accuracy can be achieved by replacing the divider

with a dedicated SoC measurement unit based on Coulomb counting. However, for simplicity and cost-effectiveness, the resistor divider method is used here.

Temperature monitoring is implemented using 10k NTC thermistors, with one thermistor attached to each cell with thermal paste or adhesive to ensure proper thermal contact. These sensors interface with the microcontroller for real-time temperature tracking. The INA219 sensor is connected to the battery pack's main terminal to measure overall current during charging and discharging, enabling efficient performance monitoring.

Software

The software is developed using the Arduino framework for the IndusBoard COIN V2 (ESP32-S2), utilising libraries such as WiFi.h, ESPAsyncWebServer.h, Wire.h, and the Adafruit INA219 library. The device supports both Station (STA) mode for connecting to an existing Wi-Fi network and AP mode to create its own hotspot.

An asynchronous web server hosts a real-time dashboard built with HTML, CSS, and JavaScript, allowing live monitoring without page refresh. The firmware reads cell voltages via ADC pins and current via the INA219 over I²C, processes the data, and updates the dashboard continuously for effective battery monitoring.

In the firmware section, the first step is to configure the Wi-Fi credentials (SSID and password) to enable network connectivity. The default number of cells in the battery pack is then defined in the code, allowing flexibility for 2S, 3S, or 4S configurations. After this, an embedded web interface is developed using HTML, CSS, and JavaScript to create a real-time dashboard for monitoring battery parameters.

Next, sensor-handling functions are implemented to read data from the ADC pins and the INA219 current

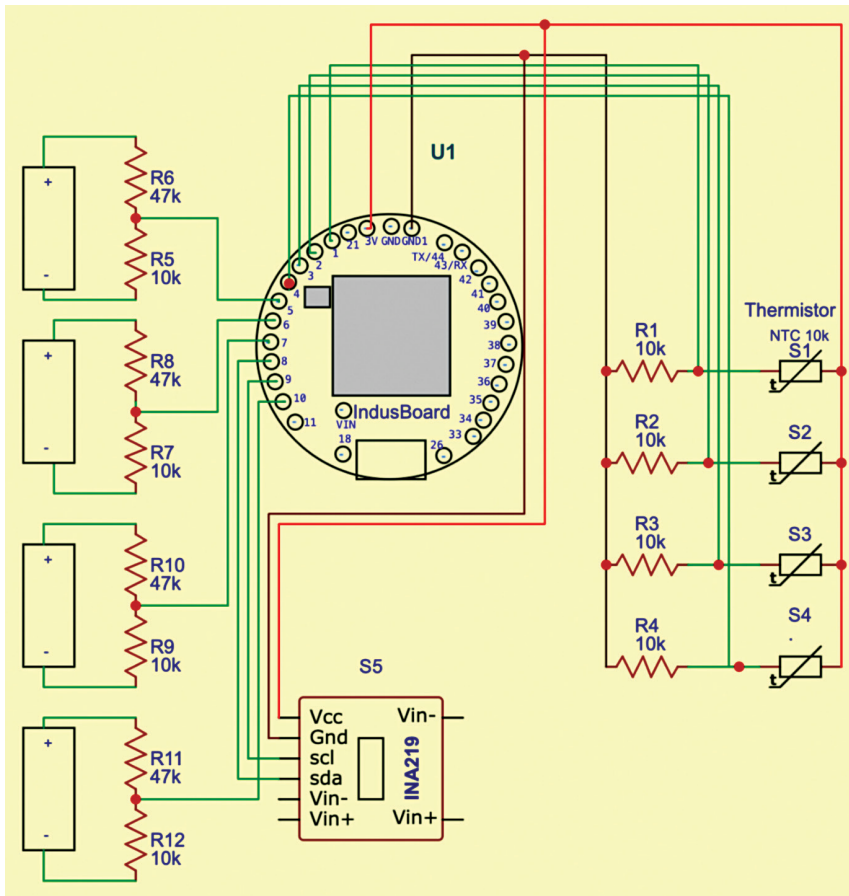


Fig. 2: Circuit diagram

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2 // IoT 4S Balanced Cell Monitor - Individual Voltage Dividers (10k + 47k)
3 // ESP32-S2 IndusBoard Coin
4 // =====
5
6 #include <WiFi.h>
7 #include <ESPAsyncWebServer.h>
8 #include <AsyncEventSource.h>
9 #include <Adafruit_INA219.h>
10 #include <Wire.h>
11
12 const char* sta_ssid = "YOUR_WIFI_SSID"; // ← CHANGE
13 const char* sta_password = "YOUR_WIFI_PASSWORD"; // ← CHANGE
14
15 const char* ap_ssid = "BatteryMonitor-AP";
16 const char* ap_password = "12345678";
17
18 #define NUM_CELLS 4
19
20 // ADC Pins for Individual Cell Voltage Dividers (10k + 47k)

```

Fig. 3: Code snippet configuring Wi-Fi